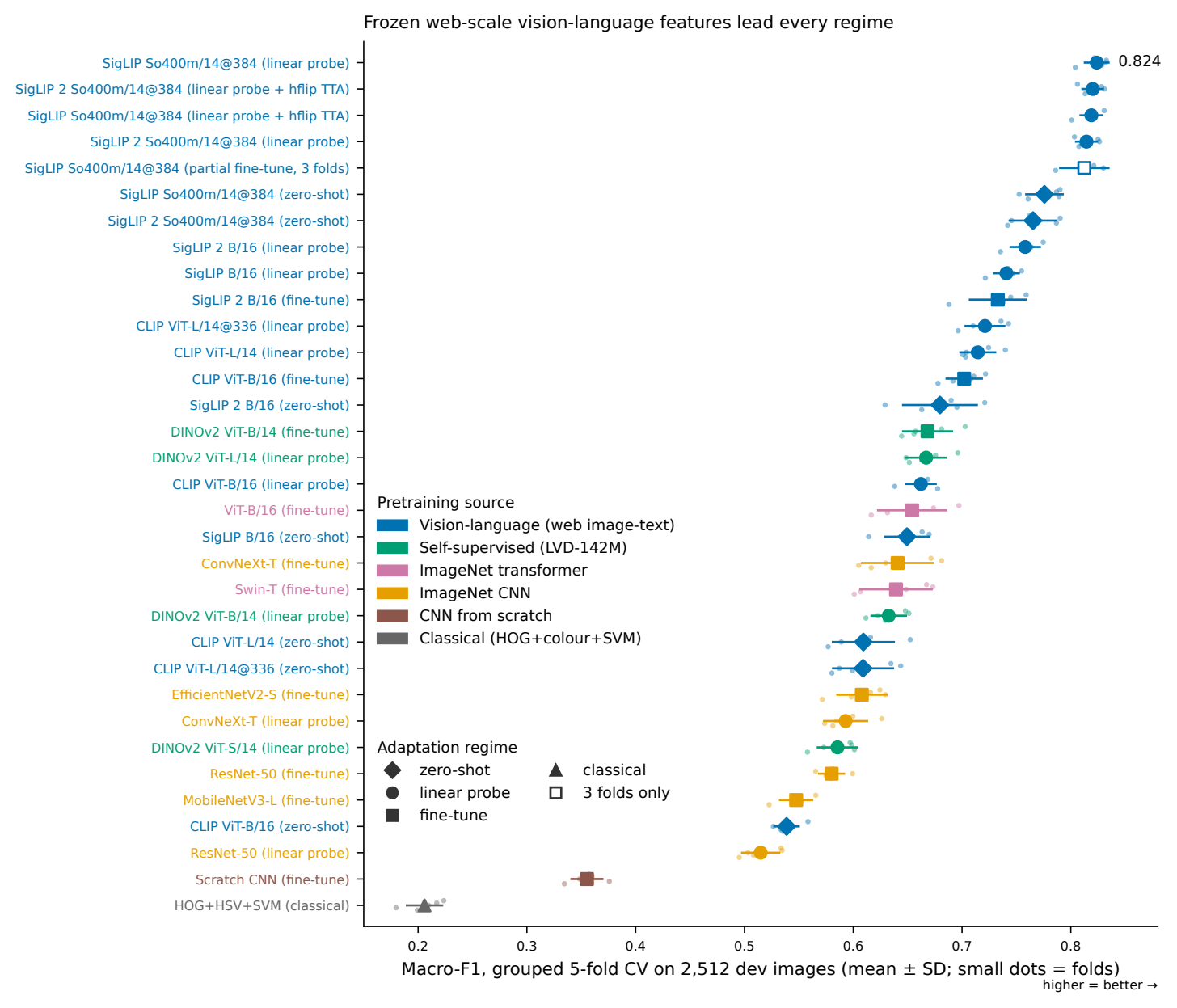
Frozen web-scale vision-language features lead every regime
SigLIP So400m/14@384 (linear probe)
SigLIP 2 So400m/14@384 (linear probe + hflip TTA)
SigLIP So400m/14@384 (linear probe + hflip TTA)
SigLIP 2 So400m/14@384 (linear probe)
SigLIP So400m/14@384 (partial fine-tune, 3 folds)
SigLIP So400m/14@384 (zero-shot)
SigLIP 2 So400m/14@384 (zero-shot)
SigLIP 2 B/16 (linear probe)
SigLIP B/16 (linear probe)
SigLIP 2 B/16 (fine-tune)
CLIP ViT-L/14@336 (linear probe)
CLIP ViT-L/14 (linear probe)
CLIP ViT-B/16 (fine-tune)
SigLIP 2 B/16 (zero-shot)
DINOv2 ViT-B/14 (fine-tune)
DINOv2 ViT-L/14 (linear probe)
CLIP ViT-B/16 (linear probe)
ViT-B/16 (fine-tune)
SigLIP B/16 (zero-shot)
ConvNeXt-T (fine-tune)
Swin-T (fine-tune)
DINOv2 ViT-B/14 (linear probe)
CLIP ViT-L/14 (zero-shot)
CLIP ViT-L/14@336 (zero-shot)
EfficientNetV2-S (fine-tune)
ConvNeXt-T (linear probe)
DINOv2 ViT-S/14 (linear probe)
ResNet-50 (fine-tune)
MobileNetV3-L (fine-tune)
CLIP ViT-B/16 (zero-shot)
ResNet-50 (linear probe)
Scratch CNN (fine-tune)
HOG+HSV+SVM (classical)
0.824
Pretraining source
Vision-language (web image-text)
Self-supervised (LVD-142M)
ImageNet transformer
ImageNet CNN
CNN from scratch
Classical (HOG+colour+SVM)
Adaptation regime
zero-shot
linear probe
fine-tune
classical
3 folds only
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Macro-F1, grouped 5-fold CV on 2,512 dev images (mean ± SD; small dots = folds)
higher = better →